1.3 Measuring and Constructing Angles

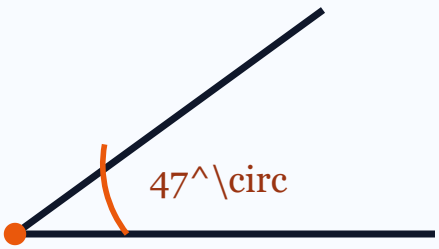
Angle measure turns a picture of two rays into a number you can compute with, the same way segment length did for two points. The Angle Addition Postulate and the definition of an angle bisector let you build equations with a variable buried inside a measure, not just read a protractor. This section also connects back to earlier ideas — opposite rays make a straight angle, and a bisector's two halves are congruent — so several problems ask you to justify a step, not only find a number.

Examples

Example 1: Classify by range

Classify $\angle A$ if $m\angle A = 47°$.

Solution



An angle less than 90° is acute.

Recall the ranges: acute is $0°$ to $90°$, right is exactly $90°$, obtuse is $90°$ to $180°$, straight is exactly $180°$.

Compare: $47°$ falls between $0°$ and $90°$.

Answer: $\angle A$ is **acute**.

Example 2: Right angle

What is the measure of a right angle, and how is it marked in a diagram?

Solution

By definition, a right angle is formed by two perpendicular rays.

A right angle is always drawn with a small square at the vertex, not a curved arc — that square *is* the symbol for 90°.

Answer: 90°.

Example 3: Straight angle

What is the measure of a straight angle, and how do its two sides relate to Section 1.1's opposite rays?

Solution

A straight angle's two sides are opposite rays — they share the vertex and point in exactly opposite directions, forming one straight line.

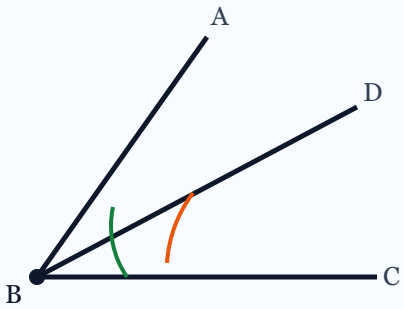
By definition, a straight angle measures exactly 180°.

Answer: 180°.

Example 4: Angle Addition Postulate

D is in the interior of $\angle ABC$. If $m\angle ABD = 35°$ and $m\angle DBC = 50°$, find $m\angle ABC$.

Solution



D is inside angle ABC, splitting it into two smaller angles.

Identify the relationship: since D is in the interior of $\angle ABC$, the Angle Addition Postulate applies: $m\angle ABD + m\angle DBC = m\angle ABC$.
Substitute: $35° + 50° = m\angle ABC$.
Answer: $m\angle ABC = 85°$.

Example 5: Find a missing part

$m\angle ABC = 110°$ and $m\angle ABD = 70°$, with D interior to $\angle ABC$. Find $m\angle DBC$.

Solution

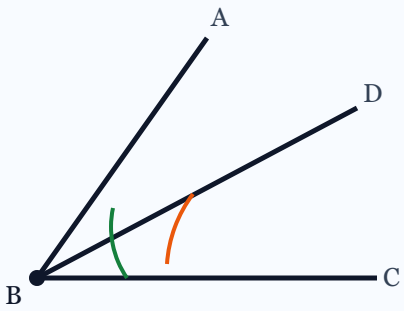
Start from the same postulate, solving for the smaller piece instead of the whole: $m\angle ABD + m\angle DBC = m\angle ABC$.
$70° + m\angle DBC = 110°$
Answer: $m\angle DBC = 40°$.

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Example 6: Angle bisector

$\overrightarrow{BD}$ bisects $\angle ABC$. If $m\angle ABD = 28°$, find $m\angle ABC$.

Solution



A bisector splits the angle into two congruent halves.

Definition: a bisector splits an angle into two *congruent* halves, so $m\angle ABD = m\angle DBC = 28°$.

$m\angle ABC = m\angle ABD + m\angle DBC = 28° + 28°$.

(Shortcut for any bisector problem: $m\angle ABC = 2 \cdot m\angle ABD$.)

Answer: $56°$.

Example 7: Bisector algebra

$\overrightarrow{BD}$ bisects $\angle ABC$. If $m\angle ABD = 3x + 2$ and $m\angle DBC = 5x - 10$, find x.

Solution

A bisector means the two halves are equal: $m\angle ABD = m\angle DBC$.

$3x + 2 = 5x - 10$

$12 = 2x \Rightarrow x = 6$.

Check: both halves equal $3(6) + 2 = 20°$. ✓

Example 8: Classify and preview supplements

Classify $\angle Q$ if $m\angle Q = 125°$, and find its supplement.

Solution

Compare to the ranges: $90° < 125° < 180°$, so $\angle Q$ is **obtuse**.

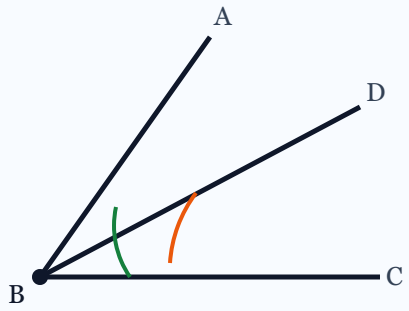
Its supplement is $180° - 125° = 55°$, which is acute — a preview of Section 1.4, where supplementary pairs are studied directly.

Answer: obtuse; supplement $55°$.

Example 9: Solve with Angle Addition

D is interior to $\angle ABC$. $m\angle ABD = 2x$, $m\angle DBC = x + 15$, and $m\angle ABC = 75°$. Find x.

Solution



The two algebraic angle pieces must add to the given whole angle.

Write the postulate as an equation: $m\angle ABD + m\angle DBC = m\angle ABC$.

$2x + (x + 15) = 75$

$3x + 15 = 75 \Rightarrow 3x = 60 \Rightarrow x = 20$.

Check: $m\angle ABD = 40°$, $m\angle DBC = 35°$, and $40° + 35° = 75°$. ✓

Example 10: Reading a protractor

A protractor's center is placed at vertex B. Ray BA lines up with the $20°$ mark and ray BC lines up with the $145°$ mark on the same scale. Find $m\angle ABC$ and classify it.

Solution

On a single protractor scale, the measure of the angle between two rays is the positive difference of their readings.

$m\angle ABC = |145° - 20°| = 125°$.

Since $90° < 125° < 180°$, the angle is **obtuse**.

Answer: $125°$, obtuse.

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Classify

Classify $m\angle A = 72°$.

Answer: **Verification:**

Problem 2: Classify

Classify $m\angle B = 90°$.

Answer: **Verification:**

Problem 3: Classify

Classify $m\angle C = 140°$.

Answer: **Verification:**

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Problem 4: Straight angle

Measure of a straight angle?

Answer: **Verification:**

Problem 5: Angle Addition

$m\angle ABD = 40°$, $m\angle DBC = 55°$. Find $m\angle ABC$.

Answer: **Verification:**

Problem 6: Find a part

$m\angle ABC = 100°$, $m\angle ABD = 65°$. Find $m\angle DBC$.

Answer: **Verification:**

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Problem 7: Bisector

$\overrightarrow{BD}$ bisects $\angle ABC$, $m\angle ABD = 32°$. Find $m\angle ABC$.

Answer: **Verification:**

Problem 8: Bisector algebra

$m\angle ABD = 4x$, $m\angle DBC = 2x + 18$, bisector. Find x.

Answer: **Verification:**

Problem 9: Solve for x

$m\angle ABD = x$, $m\angle DBC = 2x$, $m\angle ABC = 84°$. Find x.

Answer: **Verification:**

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Problem 10: Classify

Classify $m\angle D = 180°$.

Answer: **Verification:**

Problem 11: Angle Addition

$m\angle PQR = 130°$, $m\angle PQS = 80°$. Find $m\angle SQR$.

Answer: **Verification:**

Problem 12: Bisector

$m\angle ABC = 70°$, $\overrightarrow{BD}$ bisects. Find $m\angle ABD$.

Answer: **Verification:**

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Problem 13: Bisector algebra

$m\angle ABD = 5x - 4$, $m\angle DBC = 3x + 8$. Find x.

Answer: **Verification:**

Problem 14: Challenge — protractor

Ray BA reads $15°$ and ray BC reads $162°$ on the same protractor scale. Find $m\angle ABC$ and classify it.

Answer: **Verification:**

Problem 15: Challenge — double bisector

$\overrightarrow{BD}$ bisects $\angle ABC$, and $\overrightarrow{BE}$ bisects $\angle ABD$. If $m\angle EBD = 18°$, find $m\angle ABC$.

Answer: **Verification:**

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✓ Answer Key

1. Acute
2. Right
3. Obtuse
4. 180°
5. 95°
6. 35°
7. 64°
8. $x = 9$
9. $x = 28$
10. Straight
11. 50°
12. 35°
13. $x = 6$
14. 147°, obtuse
15. 72°

Great work — keep practising!